

Comparison of VADER and TextBlob labeling for sentiment analysis using machine learning and deep learning models: A study on generative AI user experience

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ABSTRACT

Acquiring awareness, understanding, and emotional engagement with generative artificial intelligence (AI) is crucial for enhancing service delivery and attracting new users. This study investigates user experiences with generative AI through comprehensive sentiment analysis. We analyzed 88,343 user comments using sentiment classification, topic modeling, and emotional analysis to examine user perceptions. Topic modeling revealed key themes shaping user attraction, including idea generation, learning, support, and task assistance. Trust in ChatGPT was found to be closely linked to perceived content quality and accuracy, with applications spanning education, productivity, programming, and general exploration. For sentiment labeling, we employed both VADER and TextBlob, with results showing that TextBlob-labeled datasets generally improved the performance of machine learning and deep learning classifiers. Among the models, the GRU achieved the highest accuracy of 96.38%, followed by the SVM at 94.54%. These findings suggest that while lexicon-based labeling methods, such as TextBlob, enhance classification effectiveness, the strongest predictive performance is achieved by advanced machine learning and deep learning models. Overall, the research highlights that user acceptance of ChatGPT is shaped not only by its practical utility but also by the emotional connections it fosters and its relevance across diverse contexts. These insights provide valuable perspectives on the dynamics of human-AI interaction.

1. Introduction

Grasping the concept of generative AI is a primary objective of service innovation. The rise of generative AI offers businesses significant benefits across various sectors, including marketing, operations, and finance, improving efficiency and optimization (Grewal et al., 2025; Holmström & Carroll, 2024). Nonetheless, it also raises societal issues such as job displacement, ethical dilemmas, and the need for responsible governance of AI systems (Papagiannidis et al., 2025; Qian et al., 2024). Generative AI, such as ChatGPT, facilitates natural conversations and provides user assistance. The latest version, ChatGPT-4, released in 2024, became the fastest-growing online service, reaching 100 million monthly active users and setting a record as the fastest-growing consumer application in history. ChatGPT's capabilities come from extensive pretraining on diverse data, making it a valuable tool for multilingual users and businesses since its release in November 2022

(Hu, 2023). Despite its popularity, especially in information searching, users report delays in customer service, which raises concerns over system quality, including data privacy, biases, and accuracy in responses, and these concerns persist. ChatGPT has also faced issues such as inadequate data protection and inaccuracies in basic math (Dwivedi et al., 2023; Kleib et al., 2024). Several studies have shown that generative AI is a powerful tool embedded with natural language, collecting data and content from platforms to predict and inform decision-making, helping analytics understand user needs (Agrawal et al., 2017; Fui-Hoon Nah et al., 2023; Nah & Ng, 2023).

The acceptability of technology often focuses on service quality; however, platforms like ChatGPT have not been thoroughly examined in the general context. In contactless services, they may struggle with a perceived lack of responsiveness, which can affect user acceptance. To bridge this gap, the present study introduces a novel approach by integrating emotional perspectives into Technology Acceptance Model

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(TAM) using an extended version (E-TAM). E-TAM builds on the traditional TAM framework by incorporating emotional elements to better understand how users accept and engage with ChatGPT as a novel technology (Dahri et al., 2024). Social media platforms, including YouTube, Instagram, TikTok, Facebook, Twitter, and Reddit, have become essential tools for communication, branding, and gathering customer insights (Pourazad et al., 2023). On the other hand, YouTube stands out as a platform where creators discuss their ChatGPT-related videos, allowing users to share their perspectives based on personal experiences.

This article aims to evaluate the performance of VADER and TextBlob labeling using machine learning and deep learning, with a focus on generative AI, and explore user acceptance and emotional evaluations of ChatGPT, particularly by examining the valuable feedback shared on YouTube channels. Emphasizing the significance of this research can inspire confidence in the audience about its impact. To accomplish this, a comprehensive range of methods will be employed to assess ChatGPT's service quality, user attitudes, technology acceptance, and service intentions. The corporate emotional perspective, integrated into the Technology Acceptance Model (TAM), will be utilized to understand better how users interact with ChatGPT through techniques such as sentiment analysis and topic modeling. Four main research questions will guide the investigation, ensuring a thorough exploration of the topic.

RQ1. What features or aspects of ChatGPT do users find the most useful or enjoyable?

RQ2. Have users ever felt frustrated or satisfied with ChatGPT's responses?

RQ3. What are the main factors affecting users' trust in ChatGPT's responses?

RQ4. What specific contexts do users typically engage with ChatGPT?

In pursuit of these research objectives, we undertook an innovative series of four comprehensive studies. Initially, we employed a distinctive lexicon-based approach that integrated two prominent methodologies: VADER and TextBlob. Following this, we subjected the sentiment labels generated by these tools to further analysis using advanced machine learning and deep learning techniques. Additionally, we implemented a diverse array of algorithms, including Stochastic Gradient Descent (SGD), Random Forest (RF), Extra Trees Classifier (ETC), Gradient Boosting Machines (GBM), Naive Bayes (NB), Support Vector Machine (SVM), Logistic Regression (LR), Decision Tree (DT), Maximum Entropy (ME), K-Nearest Neighbors (KNN), Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs), Long Short-Term Memory networks (LSTM), Bidirectional Long Short-Term Memory Network (Bi-LSTM), and Gated Recurrent Units (GRUs), to assess the efficacy of these approaches within our dataset. In the subsequent phase, we modeled the topic by employing Latent Dirichlet Allocation and Aspect-Based Sentiment Analysis. These enabled us to attain novel insights into user experiences with the ChatGPT service. Furthermore, we explored emotional sentiment through the lens of appraisal theory, which allowed us to examine the intricate emotional dynamics that shape user opinions and feelings. This approach provided a unique perspective on user satisfaction, frustration, trustworthiness, and overall acceptance of the ChatGPT service.

The present article is structured into several key sections. Section 2 presents a comprehensive literature review on generative AI and its role in digital transformation, with specific attention to user acceptance, service quality, and emotional responses. Section 3 elucidates the overall methodology framework, detailing the data collection procedures, analytical workflow, and machine learning algorithms used for sentiment and topic analysis. Section 4 reports and compares the empirical results derived from the applied models. Section 5 discusses the theoretical and practical implications drawn from the findings, while

Section 6 highlights the study's limitations and directions for future research. Finally, section 7 concludes the article by summarizing key insights and emphasizing the contribution to the field.

2. Literature review

The increasing integration of ChatGPT into daily digital interactions has sparked a growing interest in understanding the factors influencing user acceptance of this technology (Ma et al., 2025). Two influential frameworks that can guide this understanding are the SERVQUAL and Technology Acceptance Model (TAM) (Boakye et al., 2014; Taherdoost, 2018). Research has shown a strong positive correlation between perceived service quality and user acceptance, especially in service recovery situations. For instance, a study examining ChatGPT-4's response to TripAdvisor reviews found that high-quality service significantly boosted user receptiveness, particularly when addressing service failures. This highlights the importance of maintaining high service-delivery standards to enhance user acceptance. TAM has been widely recognized as a reliable predictor of user attitudes toward emerging technologies, including ChatGPT. According to TAM, perceived usefulness, which refers to how much technology enhances user performance, and perceived ease of use, which indicates how effortless it is to operate the system, play crucial roles in shaping user attitudes (Lin et al., 2007). Empirical studies involving undergraduate populations confirm that both factors significantly predict favorable attitudes toward and increased adoption of ChatGPT. Within this model, an individual's attitude summarizes their overall feelings toward the technology, shaped by their cognitive beliefs and emotional responses (Polančič et al., 2010). A positive attitude bolsters the intention to use technology, which is further influenced by social norms and contextual pressures.

The Unified Theory of Acceptance and Use of Technology (UTAUT) emphasizes the significance of user attitudes in shaping behavioral intentions (Dwivedi et al., 2019; Maruping et al., 2017; Williams et al., 2015). In addition, emotion is less explored; however, equally important aspects also play a pivotal role in technology acceptance. Drawing from appraisal theory, recent studies indicate that emotional responses, such as joy, anxiety, excitement, or frustration, mediate the relationship between cognitive evaluations and behavioral outcomes (Sander et al., 2005; Watson & Spence, 2007). The audience should recognize the significance of these emotional responses in the context of technology acceptance. These emotional responses are influenced by perceived usefulness, ease of use, and the relevance of the technology to the user's context. Notably, positive emotions can enhance user engagement and the intention to continue using AI systems. In contrast, negative emotions, especially those stemming from perceived biases or a sense of losing control, can hinder usage.

Furthermore, this research highlights the significant impact of cultural and contextual factors on user attitudes and emotional responses. This finding underscores the significance of localized design features and culturally adaptive interfaces in fostering broader acceptance among diverse user groups. It also emphasizes the relevance of the audience's cultural context in their acceptance of ChatGPT. Positive emotional reactions, such as satisfaction, gratitude, and love, commonly linked to positive experiences, foster ongoing engagement. Conversely, negative emotions such as fear, disappointment, and anger often arise from uncertainty or system failure, thereby undermining user satisfaction and trust. Additionally, how users attribute outcomes play a role in their emotional responses. Users who attribute their outcomes to their own decisions tend to be more satisfied than those who blame the AI system. Table 1 presents details of previous research related to user satisfaction with ChatGPT.

3. General outline with steps

Sentiment Analysis (SA) is a valuable tool that helps organizations understand emotions and opinions expressed in written content, such as

Table 1
Previous research on user satisfaction with ChatGPT.

Research	Research Method	Key finding
(Niu & Mvondo, 2024)	Survey	User satisfaction and loyalty toward ChatGPT and the ethical usage concern. Finding that information quality positively affects user satisfaction, perceived usefulness, and coolness. Perceiving usefulness, coolness, technology affinity, and post-human ability positively impact satisfaction.
(Duong et al., 2024)	Survey	ChatGPT's e-service quality positively influences perceived value, innovativeness and subsequent learning outcomes.
(Abdalla, 2024)	Survey	The technology acceptance positively influenced perceived usefulness and ease of use. Users valued ChatGPT's personalized help, natural interaction, and understanding of context, which boosted their intention to adopt the technology. The perception of use mainly mediated the effects of these affordances on ChatGPT acceptance.

customer reviews, social media posts, and feedback. It is crucial for monitoring brand reputation, enhancing customer support, and conducting market research. The advantages of sentiment analysis include providing real-time insights, scalability, cost-effectiveness, and better decision-making (Medhat et al., 2014). Companies can utilize this approach to enhance customer interactions, measure brand sentiment, and gain a competitive advantage. Through sentiment analysis, businesses can craft more personalized and emotionally resonant marketing strategies, leading to improved performance (He et al., 2015; Mustak et al., 2024). This process involves assessing text and reviews to determine the emotions they convey, classifying them as positive, negative, or neutral. Sentiment classification then organizes sentences according to predefined criteria (Nandwani & Verma, 2021). This technique is beneficial for analyzing social media content, product reviews, customer feedback, news articles, and product descriptions, as these text types often convey product-related sentiments and evaluations. In this study, the analysis and classification process typically involves several steps:

collecting text data, preprocessing the data, extracting relevant features, selecting those features, and finally performing sentiment analysis and classification.

3.1. Data collection and pre-processing

First, data from the top 1000 most-viewed videos related to ChatGPT users who commented on YouTube platforms were collected using a crawler-based approach via a web API, resulting in 88,343 comments. The comments vary in language and content, so pre-processing is essential to handle complexities like symbols, misspellings, emotions, stop words, and slang. Moreover, these noisy characteristics can degrade analysis performance, raising concerns about the reliability of critical processes. Fig. 1 depicts the data collection and pre-processing procedure.

3.2. Lexicon-based approach

Lexicons are collections of tokens, with each token assigned a pre-defined score indicating its neutrality, positivity, or negativity (Hung & Chen, 2016). Tokens are scored based on their polarity, such as +1, 0, and -1 for positive, neutral, and negative, respectively, or using a range from [+1, -1], where +1 represents highly positive and -1 represents highly negative, in lexicon-based approach. The scores of each token in a given text are aggregated, with positive, negative, and neutral scores summed separately. The pre-processing dataset is unlabeled, and an unsupervised approach is applied using open-source Python libraries, such as TextBlob and VADER (Valence Aware Dictionary and Sentiment Reasoner), to label. This method treats the text as a collection of words, comparing them to predefined dictionaries that provide their semantic orientation. As an unsupervised sentiment analysis technique, it effectively determines the polarity of the text, categorizing it as either positive, negative, or neutral.

3.3. TextBlob

TextBlob is a Python library designed for processing textual data, a versatile and user-friendly tool that simplifies a myriad of Natural Language Processing (NLP) tasks (Aljedaani et al., 2022). Its array of

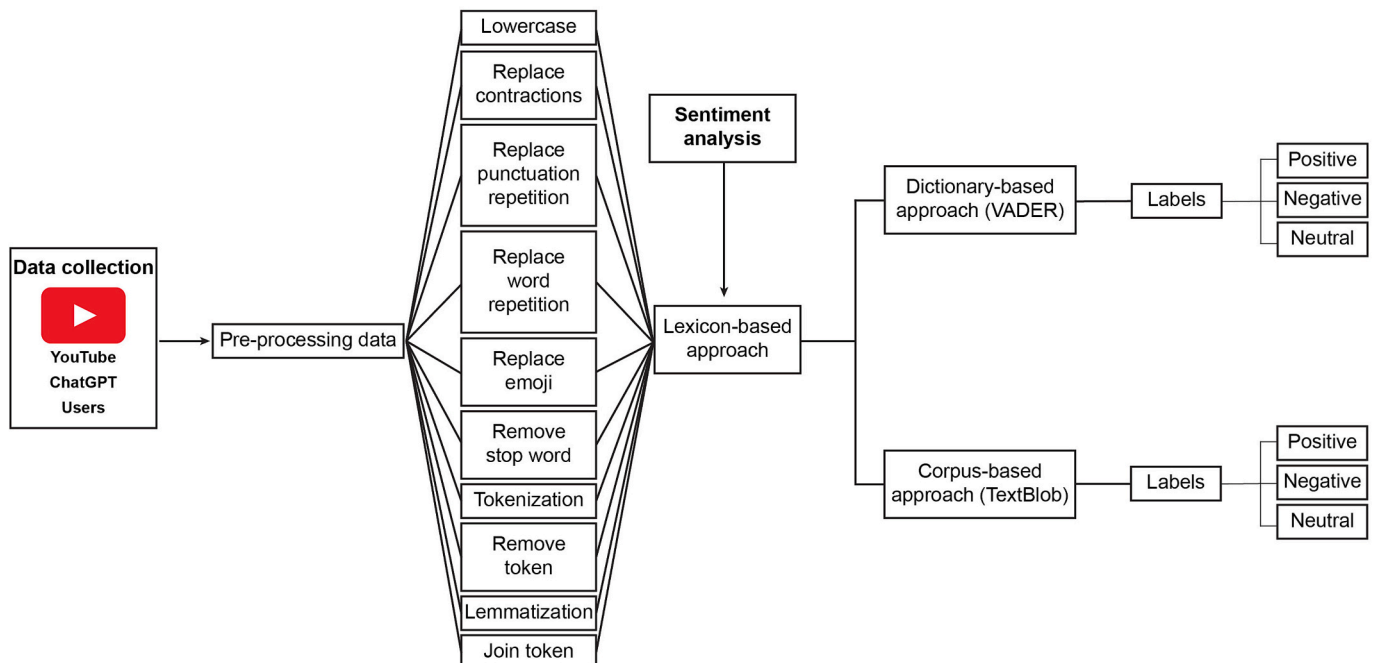


Fig. 1. Data collection and pre-processing procedure.

functions, including parts-of-speech tagging, noun phrase extraction, sentiment analysis, classification, and translation, empowers developers and data scientists to handle complex NLP projects with ease. Built on the Natural Language Toolkit (NLTK), TextBlob is a robust platform that enables practical sentiment analysis and facilitates translation of tweets across languages. Its machine learning models are designed to help users overcome the challenges of sentiment analysis. The default sentiment analyzer in TextBlob is a comprehensive tool that supports a wide range of NLP tasks, giving users the confidence to tackle their projects effectively.

3.4. Vader

Valence Aware Dictionary and Sentiment Reasoner (VADER) is a rule-based sentiment analysis tool specifically designed to analyze social media texts such as tweets, reviews, and comments (Hutto & Gilbert, 2014). It is part of the natural language toolkit library and is well-regarded for its speed and simplicity, particularly when processing short, informal texts.

3.5. Sentiment analysis

Sentiment analysis is a crucial tool for swiftly extracting insights from large amounts of text data that are highly relevant and significant in understanding users' opinions. Also known as opinion mining, sentiment analysis determines whether the opinions, views, attitudes, emotions, and feelings expressed in the text are positive, negative, or neutral. In this study, sentiment analysis is conducted using lexicon-based approaches, specifically VADER and TextBlob, for initial sentiment labeling. These labeled datasets are subsequently used to train machine learning and deep learning models for sentiment classification. We leverage natural language processing techniques to pinpoint and extract subjective information from text, thoroughly comparing algorithms to determine the most effective method for sentiment classification. In addition to sentiment classification, we integrate Latent Dirichlet

Allocation (LDA) for topic modeling and Aspect-Based Sentiment Analysis (ABSA) to gain a deeper understanding of user opinions. Fig. 2 visually represents the sentiment analysis framework used in this study.

3.6. Feature extraction

Feature extraction is crucial in sentiment classification, as it directly impacts model performance by extracting valuable information from text data. However, incorporating additional features is challenging, particularly because extracting information from text is complex. This complexity underscores the importance of feature extraction and the need for careful consideration when adding new features. This section uses Term Frequency-Inverse Document Frequency (TF-IDF) for feature extraction. TF-IDF is a statistical measure that is instrumental in determining the significance of words within a document in a dataset (Aizawa, 2003; W. Zhang et al., 2011). This function enables TF-IDF to be widely used in natural language processing applications like sentiment analysis and information retrieval. TF-IDF can be expressed as in Eqs. (1), (2), and (3):

$$TF = tf_{t,d} \quad (1)$$

$$IDF = \log\left(\frac{D}{D_t}\right) \quad (2)$$

$$TF - IDF = tf_{t,d} \bullet \log\left(\frac{D}{D_t}\right) \quad (3)$$

where $TF - IDF$ denotes the weight of term t in document d , $tf_{t,d}$ represents the term frequency of t in document d , D is the total number of documents in the corpus, and D_t is the number of documents containing term t .

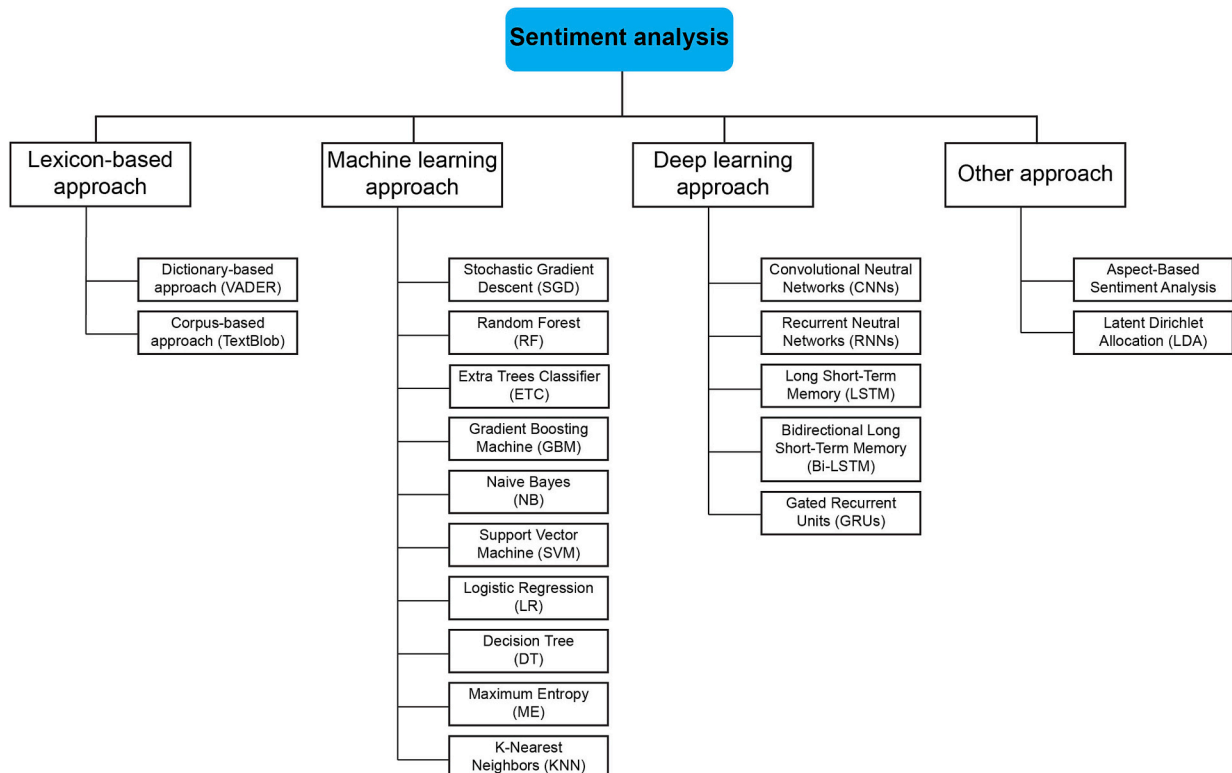


Fig. 2. Sentiment analysis approach.

3.7. Machine learning approach

3.7.1. Stochastic gradient descent

Stochastic Gradient Descent (SGD) is an optimization algorithm that approximates the gradient of the loss function using a randomly selected subset of the data and typically a single example or a mini-batch. Unlike standard (batch) gradient descent, which computes gradients using the entire dataset, SGD performs updates using only a portion, making each iteration faster and more memory-efficient. Although individual updates are noisier and less precise, the frequent updates allow the algorithm to converge more quickly, particularly in large-scale datasets. This rapid convergence makes SGD a practical choice for training deep learning and machine learning models where computational efficiency is critical. The stochastic gradient descent (SGD) algorithm significantly simplifies the process. Rather than calculating the exact gradient of each iteration, it approximates this gradient using a single randomly selected example, as expressed in Eq. (4) (Bottou, 2012; Tian et al., 2023).

$$\omega_{t+1} = \omega_t - \gamma_t \nabla_{\omega} Q(z_t, \omega_t) \quad (4)$$

where ω_t is the current model parameters iteration, γ_t is learning rate, $\nabla_{\omega} Q$ is gradient of the loss function with respect to ω_t , t is current iteration, and z_t, ω_t are a randomly selected training example.

3.7.2. Random forest

Random Forest (RF) is an ensemble machine learning algorithm designed that adapts to various scenarios. It constructs multiple decision trees during training and combines their outputs to improve predictive accuracy and control overfitting. Using a bootstrap aggregation (bagging) technique, the model draws random samples with replacements from the original dataset. A decision tree is built for each sample using a randomly selected subset of features at each split, introducing additional diversity among the trees. The final prediction is obtained by majority voting for classification tasks or averaging for regression tasks, thereby enhancing the model's generalization. By aggregating the results of multiple diverse decision trees, random forest offers robustness against noise, ensuring a secure model, improving model stability, and reduces variance compared to a single decision tree, which can be expressed as in Eq. (5) (Biau & Scornet, 2016; Breiman, 2001).

$$RF = \text{mode} \left\{ \sum_{i=1}^n T_i \right\} \quad (5)$$

where T_i is the tree in RF and n is the number of trees.

3.7.3. Extra trees classifier

The Extra Trees Classifier (ETC) is an ensemble learning method that constructs a collection of randomized decision trees to make predictions for classification problems. In contrast to random forests, extra trees incorporate an extra layer of randomness by selecting split thresholds at random for each feature. This technique generates a more varied set of trees, which helps reduce variance and enhance the model's resilience against overfitting. The decision function can be derived as in Eq. (6) (Ossai & Wickramasinghe, 2022):

$$h\{v, (i, \tau)\} = [x_i < \tau] \quad (6)$$

where v denotes the vector of features considered at the split, i is the index of the selected feature, x_i is the value of the i^{th} feature of the input sample, and $\tau \in \mathbb{R}$ is a randomly chosen threshold.

3.7.4. Gradient boosting machine

Gradient Boosting Machine (GBM) is a supervised machine learning technique that builds predictive models sequentially by minimizing errors using the gradient descent algorithm. Each subsequent model corrects the residuals from previous models, thereby significantly

enhancing prediction accuracy. The gradient boosting machine prediction can be expressed as in Eq. (7) (Friedman, 2001; Natekin & Knoll, 2013):

$$F(x) = \sum_{m=1}^M \gamma_m h_m(x) \quad (7)$$

where the gradient boosting machine $F(x)$ is a prediction for an input x , γ_m is the weight assigned to each learner, and $h_m(x)$ is the weak learners from decision tree.

3.7.5. Naive Bayes

Naive Bayes (NB) is a supervised machine learning algorithm that uses probabilities based on Bayes' theorem. It is mainly employed for classification tasks. The algorithm assumes that features are conditionally independent of each other given the class label, simplifying calculations. This assumption makes NB especially effective for handling high-dimensional datasets. By using Bayes' rule, Naive Bayes can be derived as in Eq. (8) (Frank et al., 2000; Kim et al., 2006):

$$P_{(y_i=1|x_i)} = \frac{P_{(y_i=1)} \prod_{j=1}^p P_{(x_{ij}|y_i=1)}}{\sum_{y=0}^1 P_{(y_i=y)} \prod_{j=1}^p P_{(x_{ij}|y_i=y)}} \quad (8)$$

where $P_{(y_i=1|x_i)}$ is estimate probability, y_i is a class label, x_i is features, and $P_{(x_{ij}|y_i=1)}$ is the likelihood of feature y_i and x_i , $\prod_{j=1}^p P_{(x_{ij}|y_i=1)}$ is gaussian distribution.

3.7.6. Support vector machine

A Support Vector Machine (SVM) is a robust supervised machine learning algorithm that excels at classification and regression. By identifying the optimal hyperplane, SVM maximizes the margin between distinct classes, ensuring a clear separation of data groups. This capability makes SVM an invaluable tool for achieving accurate and reliable predictions in complex datasets. The SVM classification is expressed as in Eq. (9) (Mammone et al., 2009; Suthaharan, 2016):

$$f(x) = \text{sign} \left(\sum_{i=1}^n \alpha_i y_i K(x_i, x) + b \right) \quad (9)$$

where $f(x)$ is the decision function of SVM classification, α_i represents the Lagrange multipliers, y_i is class labels, x_i is support vectors, $K(x_i, x)$ is the kernel function, and b is the bias.

3.7.7. Logistic regression

Logistic Regression (LR) is a widely used supervised machine learning algorithm primarily used for binary classification. Its fundamental principle is to model the relationship between a dependent binary variable and one or more independent variables by estimating probabilities using the simple and effective logistic (sigmoid) function. This function maps any real-valued number into the range of 0 to 1, making it particularly suitable for cases where we wish to predict the probability of a particular class membership, which can be expressed as in Eq. (10) (LaValley, 2008).

$$P_{(y_i=1|x_i)} = \frac{\exp(x_i^T \beta^*)}{1 + \exp(x_i^T \beta^*)} \quad (10)$$

where $\{y_i, x_i\}_{i=1}^n$ is an independent and identically distributed samples, $x_i = (1, x_{i1}, \dots, x_{ip})^T$ and $y_i \in \{0, 1\}$, $\beta^* = (\beta_0^*, \beta_1^*, \dots, \beta_p^*)^T \in \mathbb{R}^{p+1}$ with β_0^* being the intercept, and β_j^* being the coefficient corresponding the j^{th} predictor.

3.7.8. Decision trees

Decision Trees (DTs) are a class of supervised machine-learning algorithms used for classification and regression tasks in data analysis.

These models systematically represent decisions and their corresponding outcomes in a tree-structured format. In this architecture, the internal nodes correspond to tests performed on various dataset features, while the branches signify the outcomes of these tests. Terminal or leaf nodes indicate class labels in classification contexts or numerical values in regression contexts. This structured representation facilitates a clear interpretation of the decision-making process and enhances the model's ability to handle various predictive tasks effectively, which can be expressed in Eq. (11) (De Ville, 2013; Kingsford & Salzberg, 2008).

$$f(x) = \sum_{m=1}^M P_m I(x \in R_m) \quad (11)$$

where $f(x)$ represents a series of decision-making, P_m is the predicted value for region R_m , I is the indicator function, and M represents the number of leaf nodes.

3.7.9. Maximum entropy

Maximum Entropy (ME) is a probabilistic supervised learning method and a powerful tool for classification. It identifies the probability distribution with maximum entropy, ensuring the most uniform distribution consistent with the specified constraint, often expressed in terms of feature expectations. One of its key strengths is its ability to model the conditional class probability effectively, instilling confidence and trust in its results. The maximum entropy classifier estimates can be determined as in Eq. (12) (Harremoës & Topsøe, 2001; Matzkin et al., 2025):

$$P_{(y|x)} = \frac{1}{Z(x)} \exp \left(\sum_i \lambda_i f_i(x, y) \right) \quad (12)$$

where $P_{(y|x)}$ represents estimate probability, $Z(x)$ is the normalization factor, λ_i is parameter, and $f_i(x, y)$ is the feature functions.

3.7.10. K-nearest neighbors

K-Nearest Neighbors (KNN) is a foundational algorithm in supervised machine learning, skillfully addressing classification and regression challenges. This method relies on the principle of assessing the proximity of new data instances to existing observations, categorizing them according to the predominant class or computing the mean outcome based on the 'k' closest points in the multidimensional feature space. The classification of KNN can be expressed as in Eq. (13) (Kramer, 2013):

$$y = \operatorname{argmax}_c \sum_{x_i \in N_k(x)} I(y_i = c) \quad (13)$$

where y is the decision, $N_k(x)$ is the set of the k-nearest neighbors of the input x , y_i is the class label of neighbor x_i , I represent an indicator function.

3.8. Deep learning approaches

3.8.1. Convolutional neural networks

Convolutional Neural Networks (CNNs) have significantly transformed computer vision, particularly in processing grid-like data such as images and videos. These deep learning algorithms excel in automatically learning hierarchical spatial features through multiple layers of convolutional and pooling operations. A typical CNN architecture includes convolutional layers that apply filters to detect various features, pooling layers that reduce dimensionality while preserving essential information, and fully connected layers for classification. This structure enables CNNs to capture intricate patterns and relationships, making them highly effective for image classification, object detection, and segmentation tasks. The convolution operation in CNNs can be mathematically expressed as in Eq. (14) (Gu et al., 2018; Yamashita et al., 2018):

$$(f \bullet g)(x, y) = \sum_i \sum_j f(i, j) g(x - i, y - j) \quad (14)$$

where f is the input parameter, i, j are the position, g is the convolution kernel filter, and x, y are the convolution position feature after mapping.

3.8.2. Recurrent neural networks

Recurrent Neural Networks (RNNs) are robust neural networks designed to process sequential or time-series data, such as text, speech, video, and sensor data. Unlike feedforward neural networks, such as convolutional neural networks (CNNs), recurrent neural networks (RNNs) maintain an internal state or memory that enables them to capture dependencies within sequences over time, making them ideal for applications in natural language processing, speech recognition, and more. The mathematical expression of the RNN can be expressed as in Eqs. (15) and (16) (Medsger & Jain, 2001):

$$h_t = \sigma_h(W_{hh}h_{t-1} + W_{xh}x_t + b_h) \quad (15)$$

$$y_t = \sigma_y(W_{hy}h_t + b_y) \quad (16)$$

where h_t is the hidden state at time step t , σ_h, σ_y are the activation functions, x_t, y_t are the input and output at time step, respectively, and W_{hh}, W_{xh}, W_{hy} are the weight matrices.

3.8.3. Long short-term memory

Long Short-Term Memory (LSTM) networks are a specialized form of recurrent neural networks (RNNs) designed to address the challenges of long-range dependencies in sequential data. By mitigating the vanishing gradient problem that plagues traditional RNNs, LSTM networks can retain information over extended periods, thereby enhancing the model's ability to learn from temporally distant data points. The LSTM unit, including the forget gate, input gate, output gate, and cell state can be expressed as in Eqs. (17), and (18), respectively (Staudemeyer & Morris, 2019; Van Houdt et al., 2020):

$$y^{out_j}(t) = f_{out_j} \left(\sum_u \omega_{out_j u} y^{u(t-1)} \right); y^{in_j}(t) = f_{in_j} \left(\sum_u \omega_{in_j u} y^{u(t-1)} \right) \quad (17)$$

and

$$net_{c_j}(t) = \sum_u \omega_{c_j u} y^{u(t-1)} \quad (18)$$

where the j^{th} memory cell is denoted by c_j , which c_j is a multiplicative unit of out_j and in_j , $y^{out_j}(t)$ and $y^{in_j}(t)$ are in j 's activation at time t , and $\omega_{out_j}, \omega_{in_j}, \omega_{c_j}$ is a recurrent self-connection.

3.8.4. Bidirectional long short-term memory

Bidirectional Long Short-Term Memory (Bi-LSTM) networks represent an advanced variant of traditional extended short-term memory networks, specifically engineered to capture contextual information in sequential data better. Processing input data in forward and backward directions effectively integrates representations derived from past and future sequences. This dual processing capability provides a comprehensive understanding of the context surrounding each element in the sequence, thereby improving the performance of tasks such as natural language processing, speech recognition, and time series analysis. The architecture's ability to mitigate the limitations inherent in traditional unidirectional LSTMs makes it a powerful tool for sequence modeling tasks. The input sequence of $X = (x_1, x_2, x_3, \dots, x_t, \dots, x_T)$, which the forward ($\vec{h}_t$) and backward ($\overleftarrow{h}_t$) of the sequence related to the input time step t can be expressed as in Eqs. (19) and (20) (X. Zhang et al., 2018):

$$\vec{h}_t = LSTM_{forward}(x_t, \vec{h}_{t-1}) \quad (19)$$

$$\overleftarrow{h}_t = LSTM_{backward}(x_t, \overleftarrow{h}_{t-1}) \quad (20)$$

3.8.5. Gated recurrent units

Gated Recurrent Units (GRUs) are a robust advancement in recurrent neural networks, simplifying the architecture of long short-term memory units and effectively addressing the challenges faced by traditional RNNs. By harnessing GRUs' strengths, we can achieve better performance and efficiency in processing sequential data by using update gate ($\mathcal{Z}_t$) and reset gate (r_t), which can be derived as in Eqs. (21) and (22) (Chung et al., 2014; Dey & Salem, 2017):

$$\mathcal{Z}_t = \sigma(W_{\mathcal{Z}}x_t + U_{\mathcal{Z}}h_{t-1} + b_{\mathcal{Z}}) \quad (21)$$

$$r_t = \sigma(W_r x_t + U_r h_{t-1} + b_r) \quad (22)$$

where σ is sigmoid activation function [0,1], $W_{\mathcal{Z}}$, W_r are weight matrices for input, h_{t-1} is hidden state from previous time step, and $b_{\mathcal{Z}}$ is bias vectors.

3.9. Mapping machine learning and deep learning models sentiment analysis

Fig. 3 illustrates a workflow for sentiment analysis utilizing machine learning and deep learning models. We performed the test sequentially. In other words, we first performed dataset labeling from a lexicon-based approach by a dictionary-based approach (VADER) and a corpus-based approach (TextBlob) in order to obtain positive, negative, and neutral from the users' comments, then feature extraction was performed with map labels as: "positive: 1", "negative: 0", and "neutral:2". Subsequently, TF-IDF was implemented with the following parameters: maximum feature values of 15,000, embedding dimension of 100, training size of 80%, testing size of 20%, and random state value of 42. Moreover, machine learning models, including SGD, RF, GBM, NB, SVM, LR, DT, ME, and KNN, were evaluated, yielding accuracy, precision, recall, and F1-score results. Similarly, deep learning models were trained on the dataset using labeling and feature extraction, as in machine learning. Next, a layer represents dense embedding, and two layers represent the neural network's hidden layer. After that, the first

hidden layer used a dropout value of 0.5, and the second used a dropout value of 0.5, with a learning rate of 0.0001, a batch size of 32, and 20 epochs. The use of dropout was a strategic decision to reduce processing time while maintaining high-accuracy performance. Therefore, deep learning models, including CNN, RNN, LSTM, Bi-LSTM, and GRU, were trained and evaluated using accuracy, precision, recall, F1-score, validation loss, and validation accuracy.

3.10. Other approaches

3.10.1. Aspect-based sentiment analysis

Aspect-Based Sentiment Analysis (ABSA) offers fine-grained sentiment analysis, providing actionable insights into opinions regarding specific features. Supposedly, $S = [w_1, w_2, \dots, w_n]$ comprising n words, with an aspect term represented as a sequence of m consecutive words $[w_i, w_{i+1}, \dots, w_{i+m-1}]$, the objective of the task is to identify the sentiment polarity, such as positive, negative, and neutral, associated with each aspect term within the sentence (Do et al., 2019; Nazir et al., 2020).

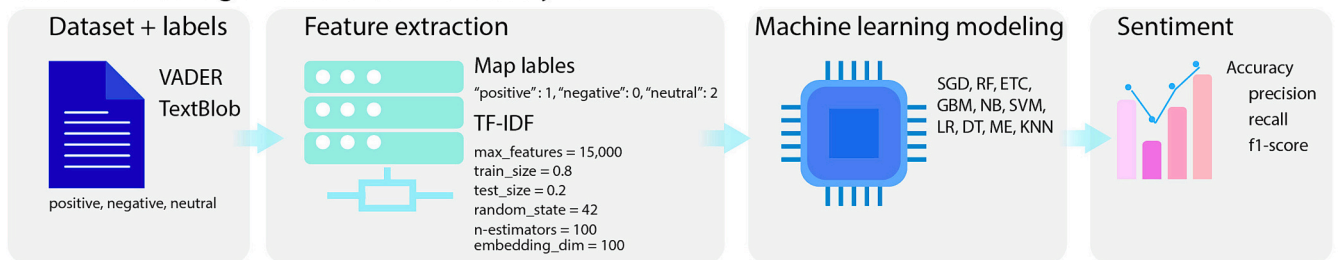
3.10.2. Latent Dirichlet allocation

Latent Dirichlet Allocation (LDA) is an unsupervised topic modeling technique widely utilized in natural language processing. This method facilitates the identification of latent thematic structures within extensive collections of text documents. By automatically discovering underlying topics and their associated words, LDA plays a crucial role in capturing the associations between documents and respective issues, thereby providing valuable insights into the distribution and nature of themes in large text corpora (Blei et al., 2003; Jelodar et al., 2019).

3.11. Performance metric

The performance of the machine learning, deep learning, and transformer-based models is evaluated using metrics such as accuracy, precision, recall, and the F1 score (Chicco & Jurman, 2020; Goutte & Gaussier, 2005; Yacouby & Axman, 2020). The accuracy can be calculated as in Eq. (23):

Machine learning models sentiment analysis



Deep learning models sentiment analysis

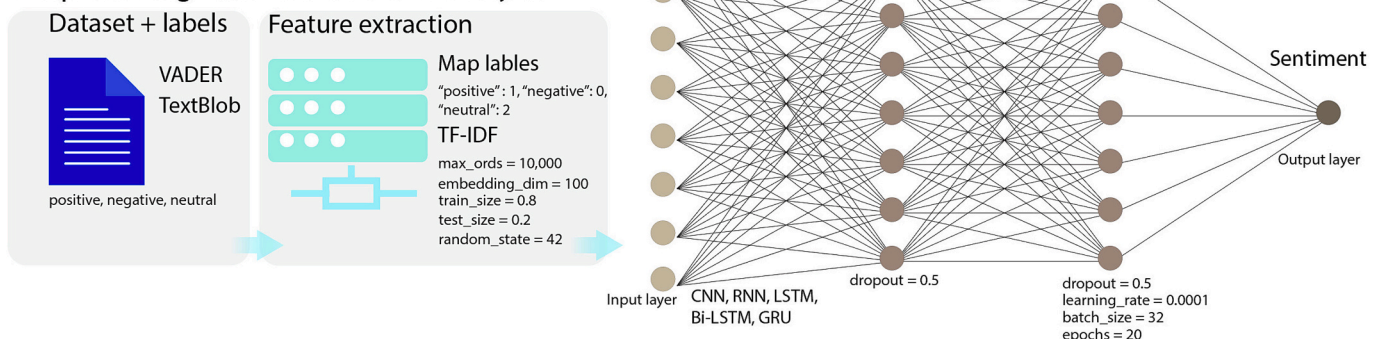


Fig. 3. Workflow of sentiment analysis using machine learning and deep learning models.

$$\text{Accuracy} = \frac{TP + TN}{(TP + TN + FP + FN)} \quad (23)$$

where, *TP* stands for true positive, *TN* for true negative, *FP* for false positive, and *FN* for false negative.

Precision is another metric used to evaluate model performance, defined as the ratio of true positives to the total number of optimistic predictions, and can be expressed as in Eq. (24):

$$\text{Precision} = \frac{TP}{(TP + FP)} \quad (24)$$

Recall is also used to evaluate model performance. It is calculated by dividing the number of true positives by the sum of true positives and false negatives, which can be expressed as in Eq. (25):

$$\text{Recall} = \frac{TP}{(TP + FN)} \quad (25)$$

The F1 score is a more suitable metric in situations with imbalanced classes because it considers both precision and recall, offering a more comprehensive assessment of the model's performance can be expressed as in Eq. (26):

$$\text{F1 Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (26)$$

4. Results

This section presents the results of our comprehensive comparative study of various sentiment analysis models, including a wide range of machine learning and deep learning models. We divided each dataset into training and test subsets, allocating 80% to training and 20% to testing. The performance of each model was evaluated using a set of comparative metrics, including accuracy, precision, recall, F1-score, and specificity, as well as validation accuracy and validation loss for deep learning models.

4.1. Sentiment labeling

Fig. 4 illustrates sentiment analysis data, categorized into positive, negative, and neutral comments using VADER and TextBlob. Table 2 details VADER and TextBlob sentiment classification from data collection. Specifically, VADER generated a larger proportion of positive comments, 56,925, compared to TextBlob's 53,795, suggesting that

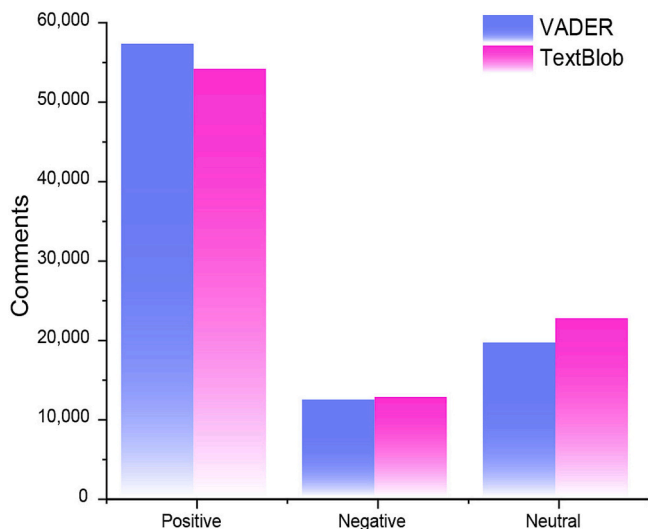


Fig. 4. Comparison of positive, negative, and neutral sentiment classification using VADER and TextBlob.

Table 2

Sentiment classification results using VADER and TextBlob.

	Positive	Negative	Neutral
VADER	56,925	12,118	19,300
TextBlob	53,795	12,161	22,387

VADER is more sensitive to positive sentiment cues. In contrast, TextBlob identified a higher proportion of neutral cues (22,387) and slightly more negative comments (12,161) compared with VADER, which detected 19,300 neutral and 12,118 negative comments. These variations highlight differences in the underlying lexicon and rule-based heuristics of the two classifiers. VADER tends to amplify affective polarity, while TextBlob provides a more balanced distribution across neutral and negative sentiments. This divergence underlines the importance of combining multiple labeling approaches to obtain a more robust sentiment representation. Hence, the outputs from VADER and TextBlob served as the foundation for subsequent analyses using machine learning and deep learning models, ensuring that the complementary strengths of these lexicon-based approaches were effectively captured in the dataset.

4.2. Machine learning sentiment models

Fig. 5 depicts the performance accuracy of machine learning sentiment models. Among all models, support vector machine (SVM) achieved the highest accuracy, with 93.15% on VADER-labeled data and 94.54% on TextBlob-labeled data, confirming their robustness across both lexicon-based approaches. Overall, datasets labeled with TextBlob consistently outperformed those labeled with VADER, suggesting that TextBlob provides training data more compatible with machine learning classifiers.

Fig. 6 presents the results of the stochastic gradient descent (SGD) model, where both VADER- and TextBlob-labeled datasets converged. Notably, TextBlob showed lower validation loss, underscoring its advantage in training efficiency. While detailed numerical scores are provided in Tables 3 and 4, the key findings highlight the support vector machine's superior performance and TextBlob's consistent advantage over VADER. Collectively, these results indicate that the support vector machine is the most reliable machine learning method for sentiment classification in this study, and that TextBlob labeling provides slightly stronger input data for supervised learning models.

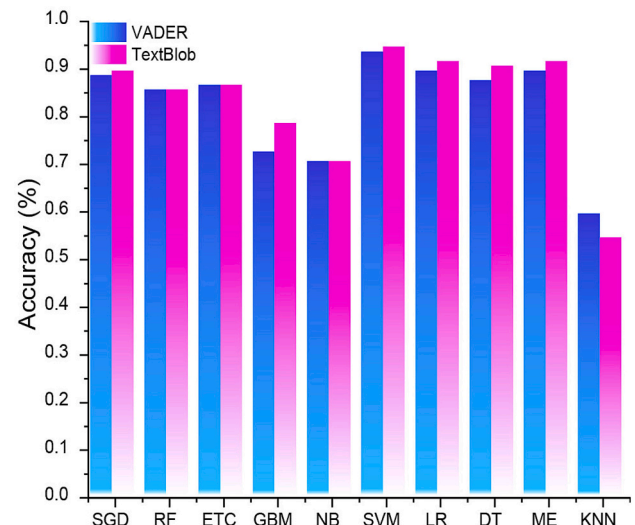


Fig. 5. Accuracy comparison among different machine learning models.

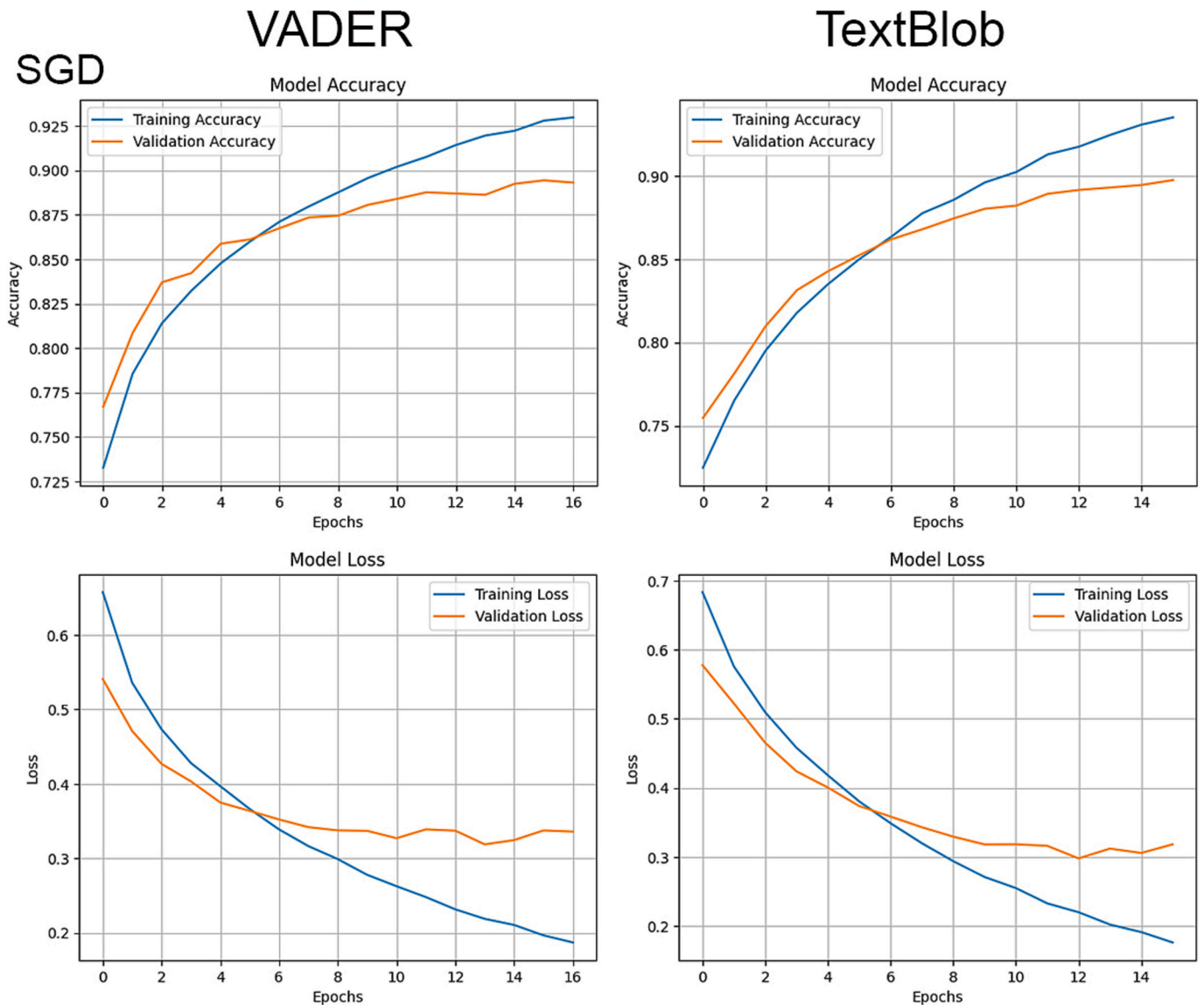


Fig. 6. Stochastic gradient descent (SGD) model performance: training and validation curves for accuracy and loss.

4.3. Deep learning sentiment models

Fig. 7 illustrates the performance accuracy for deep learning sentiment models, while Table 5 outlines their architectures. In addition, using VADER-labeled datasets, CNN, RNN, LSTM, Bi-LSTM, and GRU achieved accuracies of 94.43%, 94.46%, 94.39%, 93.85%, and **95.03%**, respectively. With TextBlob-labeled datasets, the same models achieved 95.38%, 95.83%, 95.65%, 95.67%, and **96.38%**, respectively. Among all models, gated recurrent units (GRUs) achieved the highest accuracy, reaching **95.03%** with VADER-labeled data and **96.38%** with TextBlob-labeled data, confirming their effectiveness in handling sequential text data for sentiment classification. Furthermore, across all architectures, TextBlob-labeled datasets consistently achieved slightly higher accuracies than VADER-labeled datasets, indicating that TextBlob provides more suitable training data for deep learning models. Fig. 8 presents the accuracy and loss training and validation performance of deep learning models, including CNN, RNN, LSTM, Bi-LSTM, and GRU, showing convergence for all models, with earlier stabilization across epochs. Tables 6 and 7 summarize additional metrics, including precision, recall, and F1-scores. The key findings show that GRU is the most reliable deep learning model for sentiment analysis, while TextBlob labeling often offers a clear advantage over VADER labeling. Overall, deep learning

models surpassed machine learning models in this study, demonstrating stronger capabilities for capturing contextual dependencies in user comments.

4.4. Latent Dirichlet allocation mapping

Before analyzing the data, the dataset, comprising textual comments labeled with sentiment categories, was loaded and tokenized by splitting each sentence into individual words. Latent Dirichlet Allocation (LDA) was employed to uncover user comments. Then, a Gensim dictionary in the Python library was constructed to map each unique token to a unique representation in the Bag-of-Words (BoW) format, followed by topic modeling optimized through coherence score tuning. Fig. 9 illustrates the coherence analysis as the topic numbers are varied from 1 to 24. The scores ranged from 0.46 to 0.56, with a maximum of 0.5576 at topic 4. The topic 4 model offers the most semantically meaningful and interpretable topic model for this dataset. To enhance interpretability, we employed pyLDavis, which visualizes topic distribution and term relevance through an intertopical distance map and saliency-based keyword ranking, as shown in Fig. 10. The relevance metric (λ) was set to 1 to highlight the most representative terms per topic. The analysis revealed recurring themes such as “chat,” “platform,” and “algorithm,”

Table 3
Comparison of machine learning model performance using VADER-based sentiment labels.

VADER					
Model	Sentiment	Accuracy (%)	Precision	recall	F1-score
Stochastic Gradient Descent (SGD)	Positive	88.62	0.70	0.72	0.71
	Negative		0.92	0.95	0.93
	Neutral		0.92	0.81	0.86
Random Forest (RF)	Positive	85.86	0.86	0.96	0.91
	Negative		0.90	0.40	0.56
	Neutral		0.84	0.86	0.85
Extra Trees Classifier (ETC)	Positive	86.02	0.86	0.96	0.91
	Negative		0.91	0.41	0.57
	Neutral		0.86	0.84	0.85
Gradient Boosting Machine (GBM)	Positive	72.63	0.71	0.98	0.44
	Negative		0.85	0.30	0.82
	Neutral		0.71	0.26	0.39
Naive Bayes (NB)	Positive	70.96	0.70	0.98	0.82
	Negative		0.82	0.25	0.39
	Neutral		0.85	0.19	0.32
Support Vector Machine (SVM)	Positive	93.15	0.95	0.97	0.96
	Negative		0.87	0.74	0.80
	Neutral		0.91	0.94	0.92
Logistic Regression (LR)	Positive	89.46	0.98	0.89	0.93
	Negative		0.73	0.84	0.79
	Neutral		0.80	0.94	0.87
Decision Tree (DT)	Positive	87.66	0.91	0.92	0.92
	Negative		0.66	0.59	0.62
	Neutral		0.90	0.92	0.91
Maximum Entropy (ME)	Positive	89.46	0.98	0.89	0.93
	Negative		0.73	0.84	0.79
	Neutral		0.80	0.94	0.87
K-Nearest Neighbors (KNN)	Positive	59.01	0.84	0.57	0.68
	Negative		0.61	0.38	0.46
	Neutral		0.36	0.79	0.49

Table 4
Comparison of machine learning model performance using TextBlob-based sentiment labels.

TextBlob					
Model	Sentiment	Accuracy (%)	Precision	recall	F1-score
Stochastic Gradient Descent (SGD)	Positive	89.15	0.76	0.61	0.67
	Negative		0.90	0.96	0.93
	Neutral		0.93	0.89	0.91
Random Forest (RF)	Positive	85.91	0.86	0.95	0.90
	Negative		0.93	0.40	0.56
	Neutral		0.85	0.90	0.87
Extra Trees Classifier (ETC)	Positive	86.62	0.86	0.96	0.91
	Negative		0.93	0.41	0.57
	Neutral		0.88	0.90	0.89
Gradient Boosting Machine (GBM)	Positive	78.86	0.80	0.92	0.86
	Negative		0.88	0.31	0.45
	Neutral		0.73	0.74	0.74
Naive Bayes (NB)	Positive	70.51	0.68	0.98	0.80
	Negative		0.86	0.15	0.49
	Neutral		0.86	0.35	0.49
Support Vector Machine (SVM)	Positive	94.54	0.96	0.97	0.96
	Negative		0.91	0.78	0.84
	Neutral		0.94	0.78	0.84
Logistic Regression (LR)	Positive	91.71	0.98	0.90	0.94
	Negative		0.78	0.89	0.80
	Neutral		0.86	0.98	0.92
Decision Tree (DT)	Positive	90.70	0.92	0.93	0.93
	Negative		0.70	0.68	0.69
	Neutral		0.97	0.93	0.93
Maximum Entropy (ME)	Positive	91.71	0.98	0.90	0.94
	Negative		0.78	0.89	0.83
	Neutral		0.86	0.98	0.92
K-Nearest Neighbors (KNN)	Positive	54.74	0.37	0.86	0.51
	Negative		0.61	0.29	0.40
	Neutral		0.84	0.48	0.61

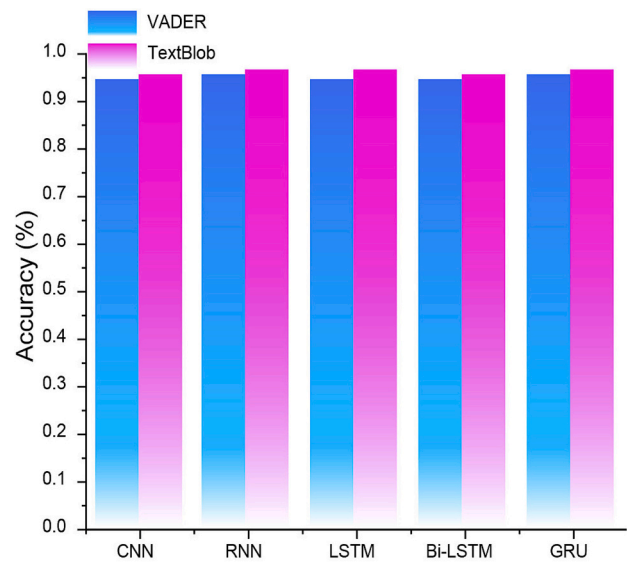


Fig. 7. Accuracy comparison among different deep learning models.

Table 5
Architecture details of the deep learning models used for sentiment analysis.

Model	Layer type	Output shape	Parameters
CNN	Embedding	Batch, 200, 100	1,000,000
	Trainable parameters:		
	Conv1D	Batch, 196, 128	64,128
	MaxPooling1D	Batch, 98, 128	0
	Dropout	Batch, 98, 128	0
	Flatten	Batch, 12544	0
	Dense	Batch, 128	1,605,760
	Dropout	Batch, 128	0
	Dense	Batch, 3	387
	Dense	Batch, 3	387
RNN	Embedding	Batch, 200, 100	1,000,000
	Trainable parameters:		
	Bidirectional	Batch, 200, 256	234,496
	Dropout	Batch, 200, 256	0
	Bidirectional	Batch, 256	394,240
	Dense	Batch, 128	32,896
	Dropout	Batch, 128	0
	Dense	Batch, 3	387
	Dense	Batch, 3	387
	Dense	Batch, 3	387
LSTM	Embedding	Batch, 200, 100	1,000,000
	Trainable parameters:		
	Bidirectional	Batch, 200, 256	234,496
	Dropout	Batch, 200, 256	0
	Bidirectional	Batch, 256	394,240
	Dense	Batch, 128	32,896
	Dropout	Batch, 128	0
	Dense	Batch, 3	387
	Dense	Batch, 3	387
	Dense	Batch, 3	387
Bi-LSTM	Embedding	Batch, 200, 100	1,000,000
	Trainable parameters:		
	Bidirectional	Batch, 200, 256	234,496
	Dropout	Batch, 200, 256	0
	Bidirectional	Batch, 256	394,240
	Dense	Batch, 128	32,896
	Dropout	Batch, 128	0
	Dense	Batch, 3	387
	Dense	Batch, 3	387
	Dense	Batch, 3	387
GRU	Embedding	Batch, 200, 100	1,000,000
	Trainable parameters:		
	Bidirectional	Batch, 200, 256	176,640
	Dropout	Batch, 200, 256	0
	Bidirectional	Batch, 256	296,448
	Dense	Batch, 128	32,896
	Dropout	Batch, 128	0
	Dense	Batch, 3	387
	Dense	Batch, 3	387
	Dense	Batch, 3	387

indicating strong discourse around AI tools and digital platforms. This approach provides both quantitative structure and qualitative insight, offering a novel perspective on user sentiment and discussion patterns. Table 8 lists the most frequent words from 24 topics in user comments. Hence, those words can be interpreted differently depending on user experience.

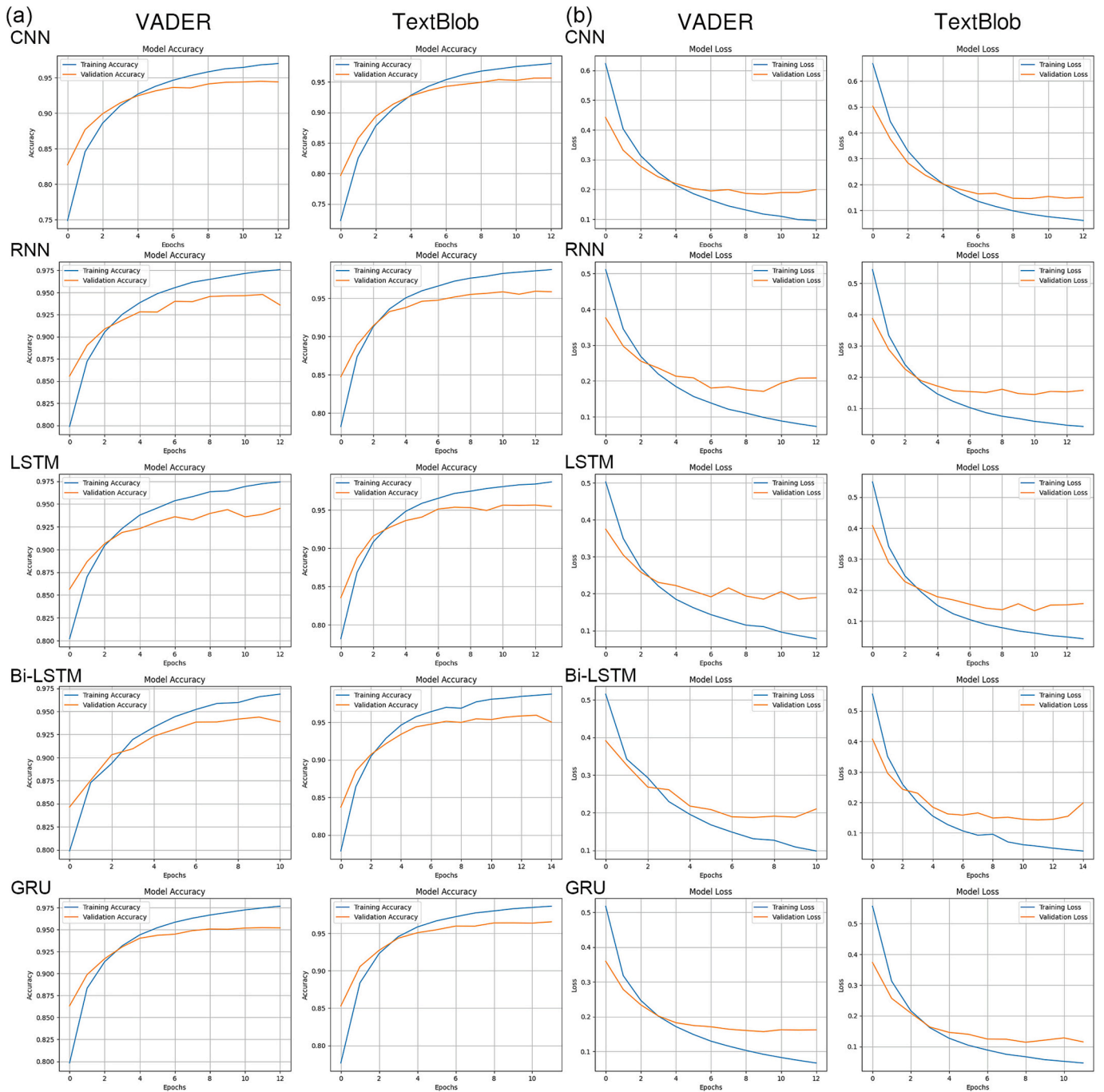


Fig. 8. Training and validation performance of deep learning models: (a) accuracy and (b) loss for CNN, RNN, LSTM, Bi-LSTM, and GRU.

4.5. Aspect-based sentiment analysis mapping

Fig. 11 depicts the aspect-based sentiment analysis from a dataset. We applied an ABSA framework to extract fine-grained insights into user perceptions by combining sentiment filtering, unsupervised clustering, and keyword extraction. Furthermore, neutral entries were removed from user comments and preprocessed via tokenization, lemmatization, and trigram generation. Then, the TF-IDF transformation was applied separately to the positive and negative subsets of the corpus, followed by k-means clustering to identify the underlying semantic grouping. Moreover, the dimensionality of these high-dimensional vectors was reduced using Principal Component Analysis (PCA) for visual interpretation, as shown in Figs. 11(a) and 11(b), which display positive and negative statements, respectively. Similarly, three distinct clusters were

identified from each sentiment category, revealing nuanced thematic structures that included both positive and negative opinions, as shown in Figs. 11(c) and (d). In addition, representative aspects within each cluster were visualized through word clouds, revealing domain-specific topics such as “content quality,” “technical support,” and “system errors.” Notably, the negative clusters emphasized practical issues (e.g., “error,” “file,” “run”) while positive clusters highlighted appreciation and learning utility (e.g., “great work,” “valuable content”). The key finding reveals that ABSA consistently identifies three domain-specific aspect clusters per sentiment. Negative feedback focuses on system reliability and usability concerns, while positive feedback emphasizes content quality and learning benefits. Hence, this unsupervised ABSA approach enabled the identification of aspect categories, offering a scalable method for opinion mining across large unstructured datasets related to

Table 6

Comparison of deep learning model performance using VADER-based sentiment labels.

VADER					
Model	Sentiment	Accuracy (%)	Precision	recall	F1-score
CNN	Positive	94.43	0.82	0.87	0.85
	Negative		0.96	0.97	0.97
	Neutral		0.96	0.91	0.94
RNN	Positive	94.46	0.87	0.85	0.86
	Negative		0.96	0.98	0.97
	Neutral		0.96	0.91	0.94
LSTM	Positive	94.39	0.87	0.83	0.85
	Negative		0.96	0.98	0.97
	Neutral		0.95	0.92	0.93
Bi-LSTM	Positive	93.85	0.86	0.82	0.84
	Negative		0.95	0.98	0.96
	Neutral		0.95	0.92	0.93
GRU	Positive	95.03	0.97	0.97	0.97
	Negative		0.84	0.90	0.87
	Neutral		0.96	0.93	0.94

Table 7

Comparison of deep learning model performance using TextBlob-based sentiment labels.

TextBlob					
Model	Sentiment	Accuracy	Precision	recall	F1-score
CNN	Positive	95.38	0.88	0.86	0.87
	Negative		0.96	0.98	0.97
	Neutral		0.97	0.95	0.96
RNN	Positive	95.83	0.90	0.88	0.89
	Negative		0.96	0.98	0.97
	Neutral		0.98	0.95	0.97
LSTM	Positive	95.65	0.89	0.87	0.88
	Negative		0.97	0.97	0.97
	Neutral		0.97	0.96	0.96
Bi-LSTM	Positive	95.67	0.87	0.89	0.88
	Negative		0.97	0.97	0.97
	Neutral		0.98	0.98	0.97
GRU	Positive	96.38	0.97	0.98	0.97
	Negative		0.91	0.89	0.90
	Neutral		0.95	0.96	0.95

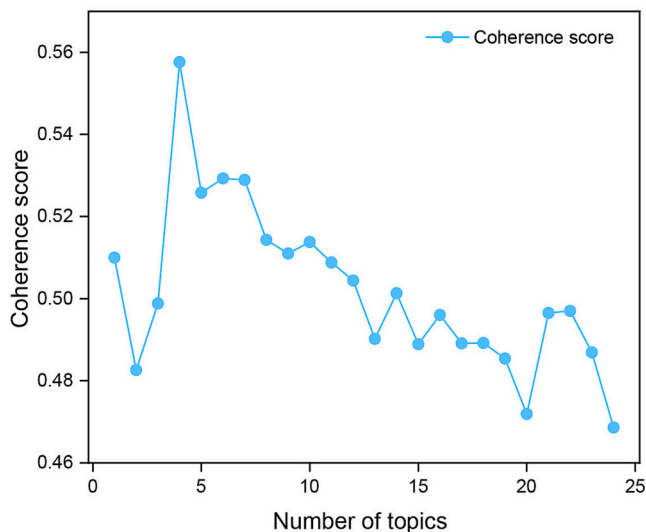


Fig. 9. Coherence vs. number of topics.

user experiences.

4.6. Emotional sentiment analysis

Emotional sentiment analysis was conducted comprehensively, identifying discrete emotional expressions in comments and categorizing them using appraisal theory to explore the emotional dynamics underlying the user's perspective. The dataset included 751 emotional words, from which 30 distinct emotions were extracted. The most frequent emotions were *happy* (42), *sad* (38), *fearful* (25), and *attracted* (21), as shown in Fig. 12. Furthermore, these emotions were then mapped to four appraisal categories: achievement, challenge, loss, and deterrence based on their underlying cognitive evaluations. For instance, “*happy*” was categorized as an achievement emotion, reflecting satisfaction or goal fulfillment, while “*fearful* and *angry*” were assigned to deterrence emotion, indicating perceived threats or external constraints. Moreover, “*attracted* and *fearless*” represented challenging emotions linked to the user's engagement with uncertainty or risk. In contrast, “*sad*” was classified under loss of emotion, corresponding to perceived damage or failure. Table 9 summarizes these appraisal theory categories. This mapping framework reveals that emotions are not merely positive or negative but are directly tied to factors influencing user trust, satisfaction, frustration, and willingness to adopt AI technologies. By leveraging appraisal theory, this approach reveals that users not only react positively or negatively but also highlight the cognitive and emotional appraisals that shape their experience with AI technologies, such as ChatGPT. For instance, this emotional sentiment analysis can be applied to improve user experience in AI chatbots by understanding and responding to users' emotional states.

5. Implications

This paper proposes a sophisticated approach for adequately collecting, monitoring, and comparing sentiment on various machine learning and deep learning algorithms using a dictionary-based approach (VADER) and a corpus-based approach (TextBlob). The comparative analysis between machine learning (ML) and deep learning (DL) models revealed that the deep learning GRU model achieved the highest accuracy of **95.03%** and **96.38%** for VADER and TextBlob, respectively. In combination with the machine learning SVM model, the results achieved the highest accuracies of **93.15%** and **94.54%** for VADER and TextBlob, respectively. Moreover, the study showed that using TextBlob labeled significantly improved ML and DL sentiment analysis. However, models like performed poorly, primarily because they are sensitive to irrelevant features and struggle with processing complex, unstructured data. Furthermore, both ML and DL models demonstrated stronger performance in classifying positive sentiments, particularly for the machine learning models, suggesting the need for more advanced feature engineering or data-balancing techniques. While deep learning models demonstrated resilience in handling these challenging classes, there remains room to improve classification precision, particularly for subtle sentiment expressions.

Overall, while deep learning models performed superiorly in handling complex text data, machine learning models, such as SVM and LR, remained competitive and could be more efficient in scenarios with limited computational resources.

Social media sentiment benchmarks were conducted by analyzing user-generated content on YouTube about ChatGPT. Multiple rounds of topic modeling generated 24 topics. After identifying the high-frequency topics, we compared the results and selected topics with the highest frequency scores related to users' perspectives.

RQ1. What features or aspects of ChatGPT do users find the most useful or enjoyable?

The features and aspects of the ChatGPT user experience are primarily appreciated for its ability to assist with idea generation, community engagement, and learning support, as reflected in the topic as follows, Topic 1: “*step*,” “*community*,” “*connect*,” “*apply*,” and “*idea*,” Topic 4: “*platform*,” “*assistant*,” “*home*,” “*video*” and “*work*,” Topic 10:

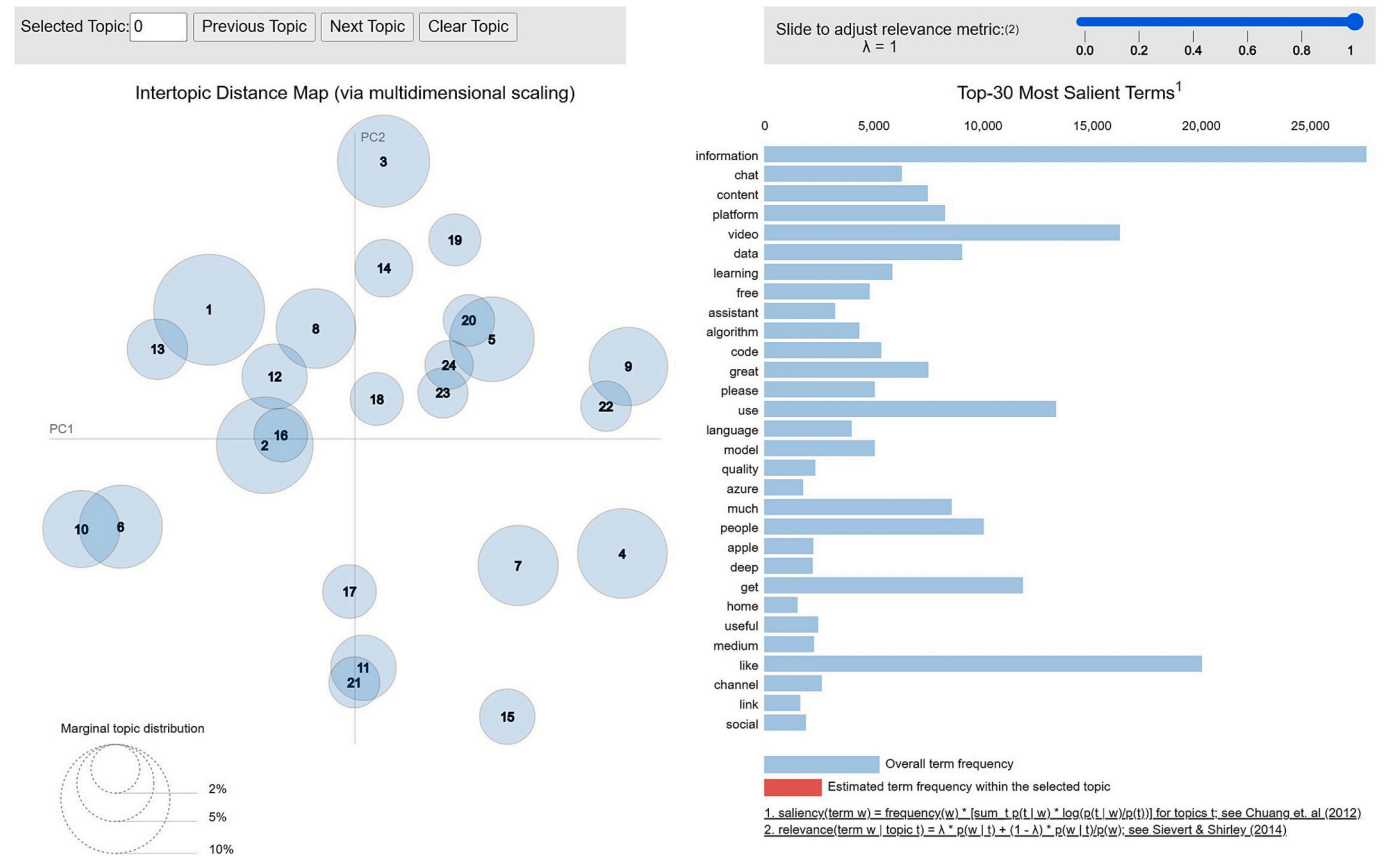


Fig. 10. Top words with high frequency mapped by Latent Dirichlet Allocation.

Table 8

Most frequency words per topic based on LDA analysis.

Topic	High frequency words in topics
1	Step, community, connect, apply, idea
2	File, error, data, code, model
3	Data, work, human, time, intelligence
4	Platform, assistant, home, video, work
5	Quit, name, information, news, extension
6	Information, people, world, country, government
7	Video, information, help, free, prompt
8	Version, install, access, script, data
9	Plus, trading, platform, price, market
10	Chat, free, use, full, developer, copilot
11	Video, algorithm, know, channel, information
12	Search, teach, people, platform, think
13	Business, potential, content, development, importance
14	Model, language, data, train, machine
15	Think, ask, use, code, something
16	Information, video, good, useful, valuable
17	Time, marketing, native, exam, intelligence
18	People, time, want, money, know
19	Research, content, student, management, discussion
20	Code, data, program, class, prompt
21	Use, information, good, people, know
22	Content, quality, social, medium, platform
23	Account, security, information, access, release
24	Quantum, twitter, mind, reality, information

“chat,” “free,” “us,” “developer,” and Topic 12: “search,” “teach,” “people,” “platform,” and “think” show that users value its accessibility, helpfulness for daily tasks, and its role as a learning companion. Moreover, Topic 16 reinforces this with terms like “information,” “video,” “good,” “useful,” and “valuable,” indicating a consistently positive evaluation of its utility. These aspects highlight that ChatGPT is viewed not only as a

technical tool but also as a practical assistant that enhances productivity and creativity. In contrast, “video” could refer to a YouTuber reviewing ChatGPT content, where the word “video” appears in user comments as a reference to the media.

RQ2. Have users ever felt frustrated or satisfied with ChatGPT’s responses?

The feedback from users on ChatGPT response is sentimentalized and classified, suggesting both satisfaction and frustration exist. Similarly, Topic 2 and Topic 8: “file,” “error,” “data,” “code,” “mode,” “version,” “install,” “access,” “script,” and “data,” indicated frustration, especially related to technical issues, script execution errors, or installation problems. Conversely, Topic 11: “algorithm,” “information,” and Topic 16: “good,” “useful,” and “valuable,” captured user satisfaction, as ChatGPT provides reliable, informative, and algorithmically competent responses. These contrasting topic clusters support the view that user experiences are highly dependent on context satisfaction, which is common in general or conceptual use. At the same time, frustration increases with technical or coding-related queries.

RQ3. What are the main factors affecting users’ trust in ChatGPT’s responses?

The key reasons to trust ChatGPT answers often correlated with its perceived accuracy, usefulness, and content quality, as well as in Topic 21 and Topic 22: “use,” “information,” “good,” “know,” “content,” “quality,” and “platform,” showed users rely on ChatGPT for dependable information and trust it for general understanding and summarization. However, hesitation arises from Topic 5 and Topic 23: “quit,” “name,” “information,” “news,” “extension,” “account,” “security,” “access,” and “release” showed that users express concerns over misinformation, personal data, and system limitations. In addition, users seem more skeptical in contexts involving security, technical failures, and ambiguous model extensions.

RQ4: What specific contexts do users typically engage with



Fig. 11. ABSA results, (a) k-means clustering of positive statements, (b) k-means clustering of negative statements, (c) word cloud of positive, and (d) word cloud of negative.

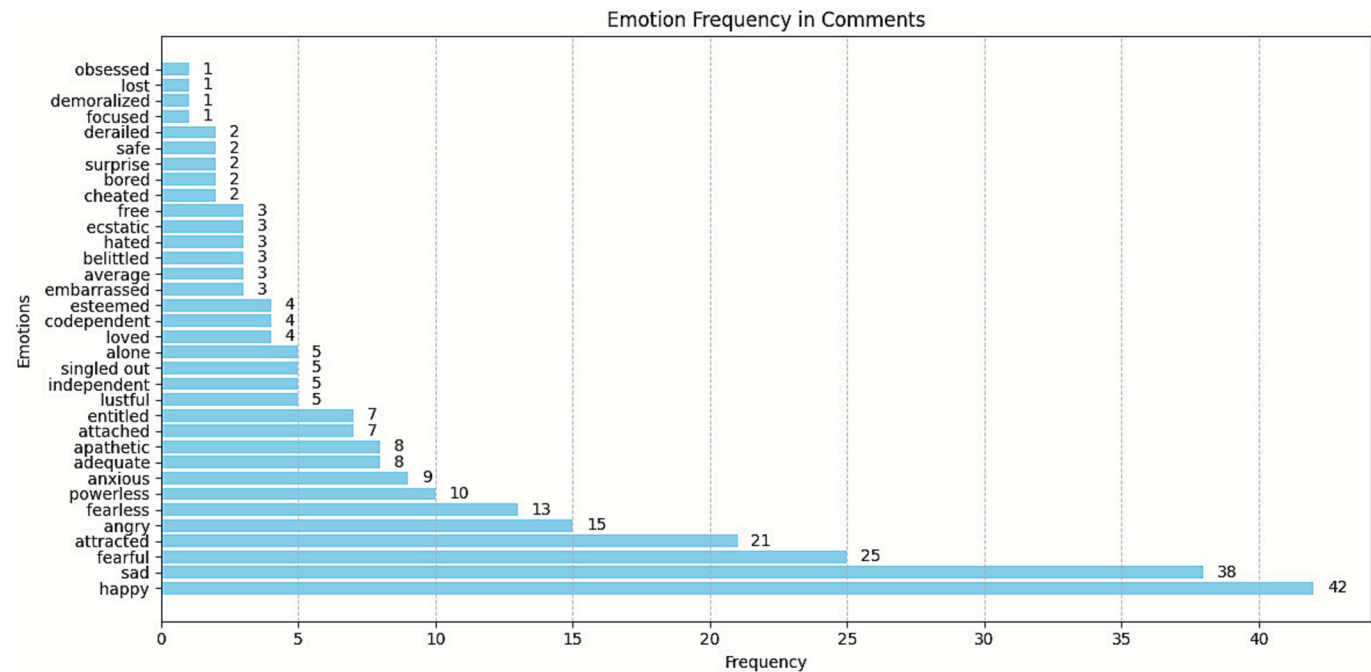


Fig. 12. Emotional distribution based on dataset analysis.

Table 9
Result of appraisal theory.

Emotion comment	Appraisal theory	Emotion
1	Achievement emotion	Happy
2	Challenge emotion	Attracted fearless
3	Loss emotion	Sad Angry
4	Deterrence emotion	Fearful

ChatGPT?

According to sentiment-classified data, ChatGPT is a versatile tool with a wide range of applications. Topics 12 and 19 demonstrate their use in study support, teaching, and academic help, while Topics 2, 15, and 20 show its relevance to coding tasks and error resolution. Topics 13 and 17 suggest their applications in business planning, marketing, and productivity. Topic 6, however, is particularly intriguing, as it showcases ChatGPT's broad exploratory uses, from global issues to advanced topics. This wide range of uses is sure to pique curiosity about ChatGPT's potential in education, learning contexts, programming, technical help, business, productivity, and general information.

Our findings reveal that sentiment analysis is not just a practical tool but also a lens into the emotional and cognitive factors that shape ChatGPT adoption. It uncovers a range of functional expectations, emotional experiences, and usage contexts that shape user interactions. Fig. 13 shows the interactions among ChatGPT users. By employing LDA-based topic modeling on sentiment-classified feedback, we found that users are primarily drawn to ChatGPT for its perceived usefulness, idea generation, educational support, and task assistance, particularly in productivity, learning, and code-generation topics. Positive sentiments reflect satisfaction and ease of use, highlighting appreciation for its accessibility and helpful responses. Negative sentiments often stem from frustration and barriers to adoption, such as technical errors, file handling issues, and price concerns. Emotional sentiment analysis further reveals that feelings of satisfaction, "happiness," and "attraction" are closely tied to topics involving helpful responses, while dissatisfaction, "fear," "anger," and "sadness" correlate with system limitations and

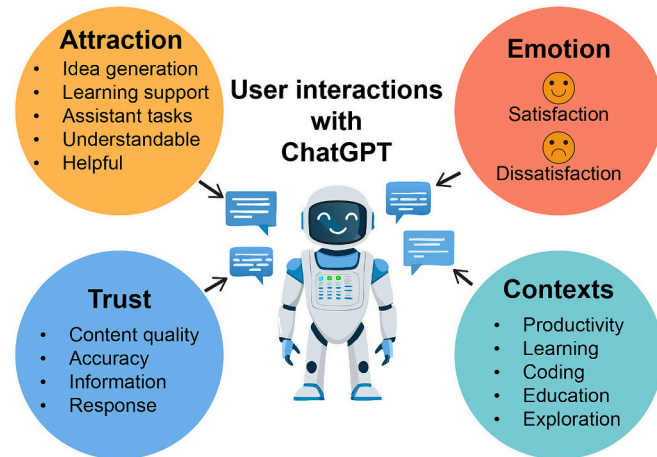


Fig. 13. Categorization of user interactions with ChatGPT.

ambiguous outputs. Trust in ChatGPT is reinforced by perceived content quality and practical accuracy, key factors in building confidence and reducing uncertainty. However, concerns about data security and misinformation remain deterrents, limiting full acceptance. In conclusion, these findings highlight that user acceptance of ChatGPT extends beyond technical functionality to encompass perceived usefulness, trust, emotional satisfaction, and reduce frustration, all central constructs in AI adoption frameworks. Strengthening these dimensions can foster greater willingness to integrate ChatGPT into diverse real-world contexts, from education to productivity.

6. Limitation and future research

This study presents several significant limitations that merit consideration. Firstly, the analysis relied solely on user comments from a single platform, which may not capture the full range of ChatGPT user experiences across various platforms and demographic groups. Additionally, while the emotional sentiment analysis highlighted specific emotions based on appraisal theory, it may not account for the more

nuanced or evolving emotional states users may experience. Future investigations could benefit from a more extensive dataset that includes information from multiple platforms and a broader range of demographic groups, thereby enhancing the applicability of the findings. Furthermore, integrating multimodal data, such as facial expressions and behavioral cues, along with testing interactions with newer iterations of ChatGPT, Gemini, DeepSeek, and Claude, could offer deeper insights into AI-user dynamics. In conclusion, by continuously refining research methodologies and expanding the analytical scope, future studies can create strategies that enhance positive emotional engagement while mitigating frustration and disengagement, ultimately paving the way for the development of more adaptive, reliable, and trustworthy AI systems.

7. Conclusion

This study analyzed ChatGPT user feedback from YouTube media using sentiment analysis and topic modeling. A comparative study of deep learning and machine learning models demonstrated that the GRU model achieved the highest accuracy in sentiment classification, while TextBlob-labeled datasets consistently yield the highest accuracy across models, reinforcing the reliability of the findings.

Topic modeling revealed overwhelmingly positive feedback regarding ChatGPT's service quality and business utility, providing reassurance and confidence in the product. While negative feedback primarily focuses on dissatisfaction with responses, ethical concerns, and technical issues, this analysis provides insights into user acceptance and identifies areas for improvement in generative AI. Emotional sentiment analysis based on appraisal theory reveals diverse responses that obtained the highest frequency of positive emotions, such as "happy," underscoring satisfaction and perceived usefulness. In contrast, negative emotions, such as "sad," "angry," and "fearful," highlight concerns about reliability, user control, and predictability. Furthermore, challenge-related emotions such as "attracted" and "fearless" indicated user engagement and openness to exploration.

Overall, these findings suggest that user acceptance of ChatGPT is shaped not only by technical performance but also by emotional dynamics and trust. Strengthening reliability, transparency, and emotional alignment will be critical for improving user satisfaction and supporting broader adoption of generative AI technologies.

CRedit authorship contribution statement

Socheat In: Writing – review & editing, Writing – original draft, Visualization, Software, Investigation, Formal analysis, Data curation, Conceptualization. **Sieb Chanchamnan:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization, Project administration.

Declaration of Generative AI and AI-assisted technologies in the writing process

Statement: The author(s) did not use generative AI technologies for the preparation of this work.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.actpsy.2026.106268>.

Data availability

Data supporting this study is available on GitHub and from the corresponding author upon reasonable request.

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